NIH NATIONAL CANCER INSTITUTE

What Are Cancer Research Studies?

What is cancer research and why is it important?

Research is the key to progress against cancer and is a complex process involving professionals from many fields. It is also thanks to the participation of people with cancer, cancer survivors, and healthy volunteers that any breakthroughs go on to improve treatment and care for those who need it.

Cancer research studies may lead to discoveries such as new drugs to treat cancer, new therapies to make symptoms less severe, or lifestyle changes to reduce the chances of getting cancer.

Cancer research may also address big picture questions like why cancer is more prevalent in certain populations or how doctors can make existing cancer detection tools more effective in health care settings.

These discoveries can help people with cancer and their caregivers live fuller lives.

Who should join cancer research studies?

When you choose to participate in a research study, you become a partner in scientific discovery. Your generous contribution can make a world of difference for people like you.

As scientists continue to conduct cancer research, anyone can consider joining a research study. The best research includes everyone, and everyone includes you.

Your unique experience with cancer is incredibly valuable and may help current and future generations lead healthier lives.

When more people of all different races, ethnicities, ages, genders, abilities, and backgrounds participate, more people benefit.

It is important for scientists to capture the full genetic diversity of human populations so that the lessons learned are applicable to everyone.

What are the types of cancer research studies?

See below for definitions on the four major types of research and their subtypes:

- basic research
- clinical research
  - clinical trials (interventional)
    - screening
    - prevention
    - treatment
    - quality of life/supportive care
  - observational
    - natural history
    - longitudinal
    - population-based
- epidemiological research
- translational research

Basic Research

Basic cancer research studies explore the very laws of nature. Scientists learn how cancer cells grow and divide, for example, by growing and testing bacteria, viruses, fungi, animal cells, and human cells in a lab. Scientists also study, for example, the genes that make up tumors in mice and rats in the lab. These experiments help build the foundation for further discovery.

Clinical Research

Why Participate

WHY IS CANCER RESEARCH IMPORTANT?

Cancer research is the key to progress against cancer. There are four major types of research: basic, clinical, epidemiological, and translational.

Basic Research
These studies allow scientists to learn how cancer cells grow and divide and build the foundation for further discovery.

Clinical Research
There are two main types of clinical research studies: **observational studies** and **clinical trials**. Observational studies collect and analyze data from people over a long period of time, while clinical trials involve treatments or care approaches that may change the results for a person with cancer.

Epidemiological Research
The study of the patterns, causes, and effects of cancer in a group of people of a certain background.

Translational Research
A research approach that moves across research disciplines, from basic lab research into clinical settings and from clinical settings into everyday care.

Clinical research involves the study of cancer in people. These cancer research studies are further broken down into two types: clinical trials and observational studies.

in a Clinical Trial?

Get information on how to evaluate a clinical trial and what questions to ask.

- **Clinical trials** are research studies that involve an intervention, which is a treatment or change that may affect the results of cancer. These can lead to new treatments, care, and improved results for people with cancer, their loved ones, and people with a high risk of cancer.
  - Treatment trials test how safe and useful a new treatment or way of using existing treatments is for people with cancer. Test treatments may include drugs, approaches to surgery or radiation therapy, or combinations of treatments.
  - Prevention trials are for people who do not have cancer but are at a high risk for developing cancer or for cancer coming back. Prevention clinical trials target lifestyle changes (doing something) or focus on certain nutrients or medicines (adding something).
  - Screening trials test how effective screening tests are for healthy people. The goal of these trials is to discover screening tools or methods that reduce deaths from cancer by finding it earlier.
  - Quality-of-life/supportive care tests aim to help people with cancer, as well as their family and loved ones, cope with side effects like pain, nutrition problems, nausea and vomiting, sleeping problems, and depression. These trials may involve drugs or activities like therapy and exercising.

Find Observation Studies >

View a studies that are looking for people now.

- **Observational studies** are studies that collect and analyze data from people. Some of these take place over a long period of time. These studies do not study the effect of a specific treatment or change.
  - Natural history studies look at certain conditions in people with cancer or people who are at a high risk of developing cancer. Researchers often collect information about a person and their family medical history, as well as blood, saliva, and tumor samples. For example, a biomarker test may be used to get a genetic profile of a person's cancer tissue. This may reveal how certain tumors change over the course of treatment.
  - Longitudinal studies gather data on people or groups of people over time, often to see the result of a habit, treatment, or change. For example, two groups of people may be

identified as those who smoke and those who do not. These two groups are compared over time to see whether one group is more likely to develop cancer than the other group.

- Population-based studies explore the causes of cancer, cancer trends, and factors that affect cancer care in specific populations. For example, a population-based study may explore the causes of a high cancer rate in a regional Native American population.

Epidemiological Research

Epidemiological research is the study of the patterns, causes, and effects of cancer in a group of people of a certain background. This research encompasses both observational population-based studies but also includes clinical epidemiological studies where the relationship between a population's risk factors and treatments are tested.

Translational Research

Translational research is when cancer research moves across research disciplines, from basic lab research into clinical settings, and from clinical settings into everyday care. In turn, findings from clinical studies and population-based studies can inform basic cancer research. For example, data from the genetic profile of a tumor during an observational study may help scientists develop a clinical trial to test which drugs to prescribe to cancer patients with specific tumor genes.

Monica Bertagnolli, Director, NIH; former director, NCI; cancer survivor

Participation in Cancer Research Matters

I am so happy to have the opportunity to acknowledge the courage and generosity of an estimated 494,018 women who agreed to participate in randomized clinical trials with results reported between 1971 and 2018.

Their contributions showed that mammography can detect cancer at an early stage, that mastectomies and axillary lymph node dissections are not always necessary, that chemotherapy can benefit some people with early estrogen receptor–positive, progesterone receptor–positive, HER2-negative breast cancer but is not needed for all, and that hormonal therapy can prevent disease recurrence.

Credit: Brigham and Women's Hospital

For just the key studies that produced these results, it took the strength and commitment of almost 500,000 women. *I am the direct beneficiary of their contributions, and I am profoundly grateful.*

The true number of brave souls contributing to this reduction in breast cancer mortality over the past 30 years? Many millions. These are our heroes.

— From NCI Director's Remarks by then-NCI Director Monica M. Bertagnolli, M.D., at the American Society of Clinical Oncology Annual Meeting, June 3, 2023

Updated: September 8, 2023

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